

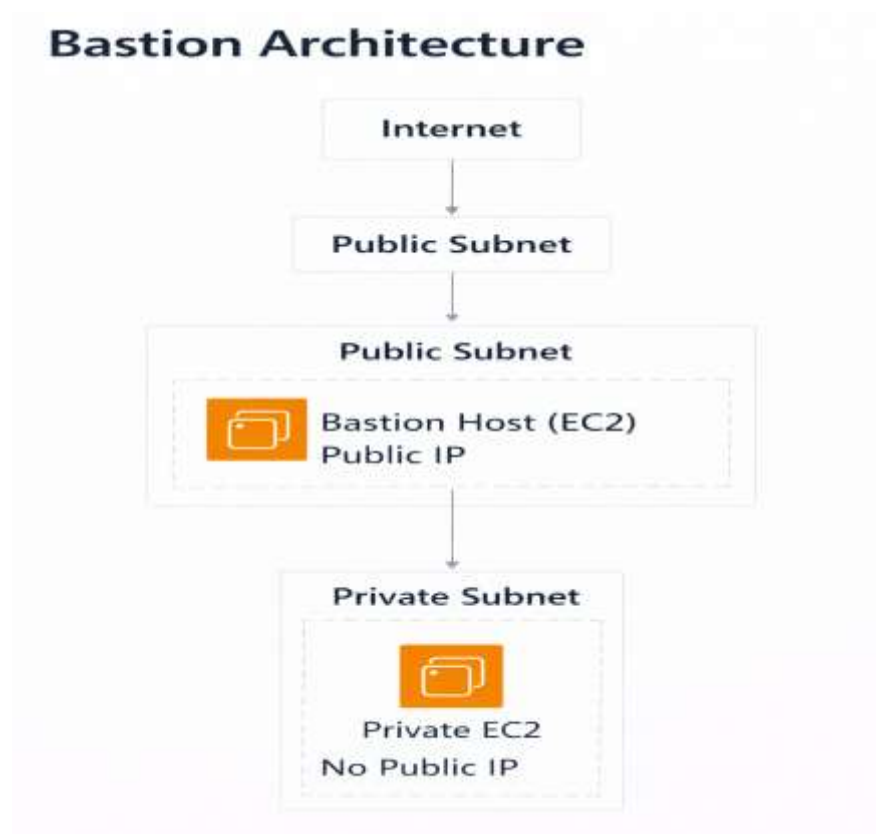
Jump Host (Bastion Host), NAT Gateway, and NAT Instance - Detailed Documentation

Jump Host / Bastion Host

What is a Jump Host (Bastion Host)?

A Jump Host (also called a Bastion Host) is a secure server placed in a public subnet that allows administrators to access private servers inside a private subnet.

It acts as an entry point to access internal resources securely.



Why Bastion Host is Used

Private subnet instances:

- Do not have public IP
- Cannot be accessed directly from the internet
- Are protected for security

So we use:

Internet → Bastion Host → Private Server

How Bastion Host Works

Step-by-step:

1. User connects to Bastion Host using SSH
2. Bastion Host connects to Private Server
3. Private servers remain hidden from internet

Bastion Host Security Rules

Bastion Security Group

Allow:

Type: SSH
Port: 22
Source: Your IP address

Private Instance Security Group

Allow SSH only from Bastion Host:

Type: SSH
Port: 22
Source: Bastion Security Group

Bastion Host Use Cases

- Secure SSH access to private servers
- Database administration
- Production environment access
- Remote infrastructure management

Real-Time Example

Company setup:

Web Server → Public Subnet
App Server → Private Subnet
DB Server → Private Subnet
Admin Access → Bastion Host

Admin Flow:

Laptop → Bastion → App Server → Database

When We Use Bastion Host

We use Bastion Host when:

1. Access Private Servers Securely

Private servers do not have public IP.

Example:

- App server in private subnet
- Database server in private subnet

Access flow:

Admin → Bastion → Private EC2

NAT Gateway

What is NAT Gateway?

A **NAT Gateway** allows **private subnet instances** to access the internet **without exposing them to incoming internet traffic**.

Private instances:

Can access internet

Cannot be accessed from internet

Why NAT Gateway is Needed

Private servers need internet access for:

- Software updates
- Download packages
- OS patches
- External API calls

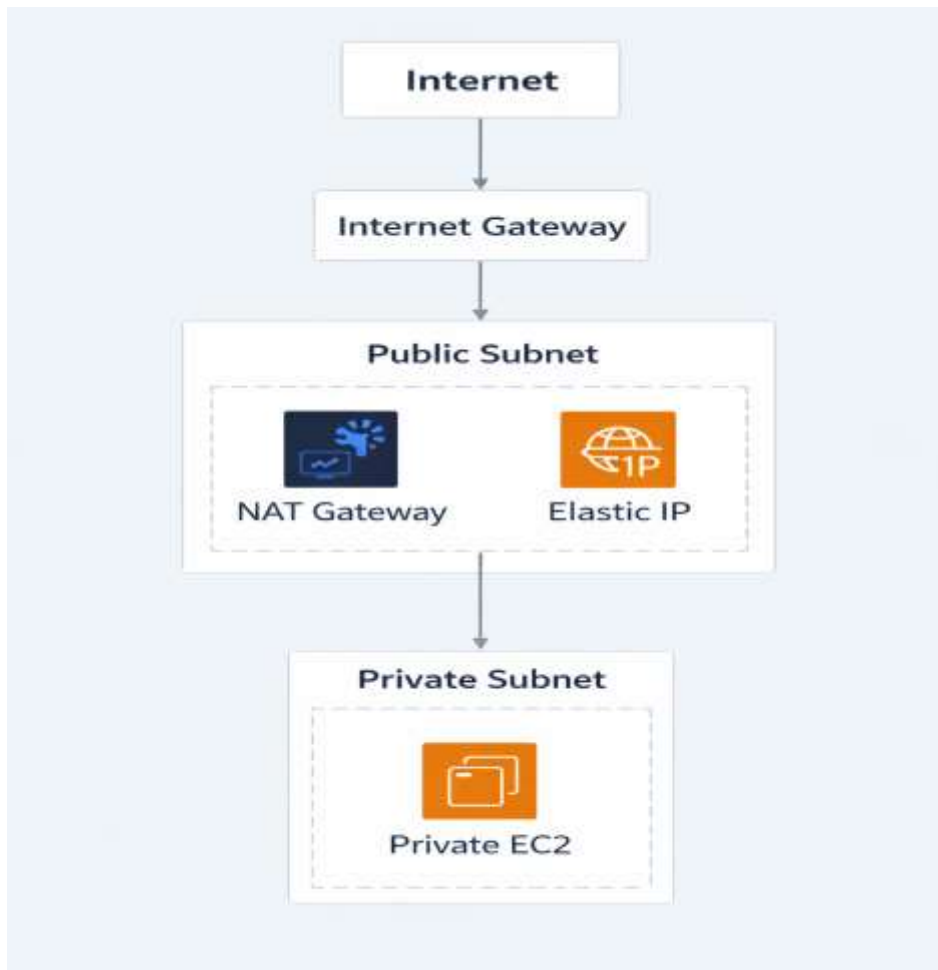
Without NAT:

apt update ✗ fails

yum install ✗ fails

wget ✗ fails

NAT Gateway Architecture



NAT Gateway Requirements

Must have:

- ✓ Public Subnet
- ✓ Internet Gateway
- ✓ Elastic IP

NAT Gateway Route Table Setup

Public Route Table

Destination: 0.0.0.0/0

Target: Internet Gateway

Private Route Table

Destination: 0.0.0.0/0

Target: NAT Gateway

NAT Gateway Advantages

- Fully managed by AWS
- Highly available
- Scalable automatically
- No maintenance required

NAT Gateway Disadvantages

- Costs money per hour
- Additional data transfer cost

When We Use NAT Gateway

We use NAT Gateway when:

1.Private Servers Need Internet Access

Private EC2 needs:

- Software updates
- Security patches
- Download dependencies

2. Secure Internet Access for Private Systems

We want:

✓ Outgoing access

✗ Incoming access

3.Production Environments (Most Common Use)

NAT Gateway is widely used in:

- Enterprise applications
- SaaS platforms
- Web applications

Typical production design:

Public Subnet:
Load Balancer
NAT Gateway

Private Subnet:
App Servers
DB Servers

When High Availability Is Required

NAT Gateway:

- ✓ Managed by AWS
- ✓ Highly reliable
- ✓ Automatically scalable

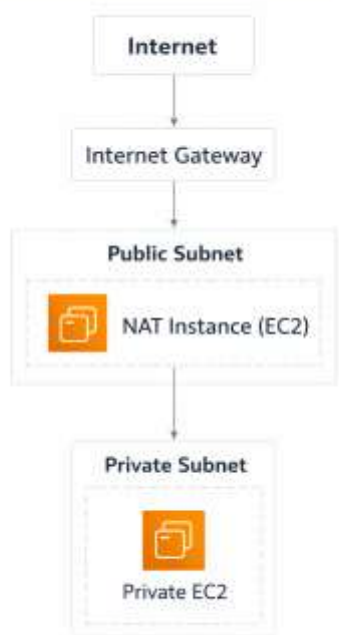
NAT Instance

What is NAT Instance?

A **NAT Instance** is an **EC2 instance configured to act as a NAT device**.

It performs the same function as NAT Gateway but is **self-managed**.

NAT Instance Architecture



NAT Instance Requirements

Must configure:

- ✓ Disable Source/Destination Check
- ✓ Enable IP Forwarding

- ✓ Configure iptables
- ✓ Assign Elastic IP

NAT Instance Advantages

- Cheaper than NAT Gateway
- Fully customizable
- Useful for small environments

NAT Instance Disadvantages

- Requires maintenance
- Manual scaling
- Single point of failure

Less reliable than NAT Gateway

When We Use NAT Instance

We use NAT Instance when:

1. Low-Cost Environments

NAT Gateway costs money hourly.

For:

- Labs
- Testing
- Small projects

NAT Instance is cheaper.

2. Custom Network Configurations

Sometimes we need:

- Custom routing rules
- Firewall rules
- Traffic inspection

NAT Instance allows customization.

3. Learning and Testing

Used in:

- DevOps labs
- AWS practice environments
- Small workloads

4. Small Traffic Workloads

If traffic is low:

- NAT Instance is sufficient
- Cost savings possible

NAT Gateway vs NAT Instance

Feature	NAT Gateway	NAT Instance
Managed by AWS	Yes	No
High Availability	Yes	No
Maintenance	None	Required
Performance	High	Medium
Cost	Higher	Lower
Scaling	Automatic	Manual

Bastion Host vs NAT Gateway

Feature	Bastion Host	NAT Gateway
Purpose	SSH access	Internet access
Used by	Admin users	Private servers
Internet access	Yes	Yes
Direct login	Yes	No